

Predicting Climate Anomalies and Emissions from Socioeconomic Indicators: A Multi-Target Machine Learning Approach

Group Members:

Abdul Moiz Hussain – 23K-0553

Muhammed Abdullah Khan – 23K-0607

Name 3 – [ID/Role]

Name 4 – [ID/Role]

Course: Data Science

Area of Specialization: Machine Learning / Environmental Data Science

Nature of Project: Technical Report / Data Science Research Project

Submission Date: 01-5-2026

CONTENTS

Abstract	3
1 Introduction	4
2 Related Work	4
2.1 Machine Learning in Climate Science	4
2.2 Socioeconomic Drivers of Carbon Emissions	4
2.3 Multi-Target Environmental Prediction	5
2.4 Interpretability in Environmental ML	5
3 Research Gap	5
4 Purpose	5
5 Objectives	6
6 Exploratory Data Analysis	6
6.1 Temporal Trends	6
6.2 Feature-Target Relationships	6
7 Methodology	8
7.1 Data Collection & Preprocessing	8
7.2 Feature Engineering	8
7.3 Modeling Framework	8
7.4 Hyperparameter Tuning	8
7.5 Evaluation & Interpretation	8
7.6 Reproducibility	8
8 Implementation	9
8.1 Technical Stack & Architecture	9
8.2 Reproducibility Measures	10
8.3 Key Implementation Snippets	10
9 Experimental Results & Evaluation	11
9.1 Evaluation Framework	11
9.2 Model Performance Comparison	11
9.3 Interpretation Results (SHAP Analysis)	12
9.3.1 Global Feature Importance	12
9.3.2 Marginal Effect Visualization	12

10 Comparative Analysis of Models	13
10.1 Accuracy vs. Complexity Trade-offs	13
10.2 Target-Specific Model Behavior	15
10.2.1 CO ₂ Emissions: Deterministic Predictability	15
10.2.2 Temperature Anomaly: Fundamental Predictability Limit	15
10.3 Practical Recommendations	15
References	16

ABSTRACT

Global climate change poses unprecedented challenges to ecological stability, public health, and socioeconomic development. While traditional climate forecasting relies on complex physical simulation models, data-driven machine learning (ML) approaches have emerged as complementary tools for identifying macro-level patterns from historical observations. This project develops a temporally validated ML pipeline to simultaneously predict temperature anomalies and CO₂ emissions using century-scale country-level socioeconomic panel data. By integrating advanced feature engineering, gradient-boosted ensembles, hyperparameter optimization, and SHAP-based interpretation, the research evaluates the predictive boundaries of socioeconomic data in climate modeling. Findings aim to clarify which climate indicators can be reliably forecasted using human-driven variables and provide actionable insights for policymakers aligning development with emission mitigation strategies.

Keywords: climate forecasting, multi-target regression, socioeconomic indicators, machine learning, SHAP interpretation, temporal validation

INTRODUCTION

Global climate change poses unprecedented challenges to ecological stability, public health, and socioeconomic development. While traditional climate forecasting relies on complex physical simulation models, data-driven machine learning (ML) approaches have emerged as complementary tools for identifying macro-level patterns from historical observations. Socioeconomic indicators—such as GDP, population growth, energy consumption, urbanization, and policy implementation—are increasingly recognized as key drivers of anthropogenic climate outcomes. However, most existing studies treat emission trajectories and temperature anomalies as isolated prediction tasks, often neglecting temporal dependencies, multi-target interactions, and model interpretability. This project addresses these limitations by developing a rigorous, temporally validated ML pipeline that simultaneously predicts temperature anomalies and CO₂ emissions using panel data spanning over a century of country-level socioeconomic records. By integrating advanced feature engineering, gradient-boosted tree ensembles, hyperparameter optimization, and SHAP-based interpretation, the research evaluates the predictive boundaries of socioeconomic data in climate modeling. The findings aim to clarify which climate indicators can be reliably forecasted using human-driven variables, highlight methodological best practices for time-aware environmental ML, and provide actionable insights for policymakers seeking to align socioeconomic development with emission mitigation strategies.

RELATED WORK

Machine Learning in Climate Science

Recent studies have applied Random Forests, neural networks, and hybrid physics-ML models to downscale climate projections and predict extreme weather events [1]. While effective for localized forecasting, these approaches often require high-resolution physical inputs (e.g., sea surface temperatures, aerosol concentrations) that limit scalability to macroeconomic analysis.

Socioeconomic Drivers of Carbon Emissions

Research grounded in the Kaya identity consistently demonstrates strong correlations between GDP per capita, energy intensity, and CO₂ emissions [2]. Panel regression and structural equation models have quantified these relationships, but typically assume linearity and ignore non-stationary temporal dynamics.

Multi-Target Environmental Prediction

Few studies simultaneously model multiple climate indicators. Multi-output regression frameworks have been explored in air quality and water resource management [3], yet temporal leakage and inadequate cross-validation remain prevalent methodological flaws.

Interpretability in Environmental ML

The adoption of SHAP and LIME has improved transparency in ecological modeling [4]. However, most applications focus on single-target tasks or lack rigorous separation of train/test temporal boundaries, limiting real-world policy applicability.

RESEARCH GAP

Despite growing interest in data-driven climate modeling, significant gaps persist:

- (1) Limited use of temporally strict validation that prevents future data leakage in panel datasets;
- (2) Scarce comparative analysis of how different ML architectures handle multi-target climate prediction;
- (3) Inadequate interpretability frameworks that disentangle driver effects across distinct climate indicators;
- (4) Insufficient evaluation of the predictive limits of purely socioeconomic features for physical climate variables.

This project bridges these gaps by implementing a time-aware, multi-output ML pipeline with explicit feature selection, hyperparameter tuning, and target-specific SHAP interpretation.

PURPOSE

The purpose of this project is to develop, evaluate, and interpret a reproducible machine learning pipeline that predicts temperature anomalies and CO₂ emissions from historical socioeconomic indicators. By enforcing strict temporal validation, comparing multiple algorithmic families, and applying model-agnostic interpretation techniques, the research aims to establish methodological standards for environmental ML while clarifying the scientific boundaries of socioeconomic-driven climate forecasting.

OBJECTIVES

1. Construct a temporally validated multi-target regression pipeline using country-level panel data (1900–2023).
2. Engineer domain-informed features (per-capita metrics, energy ratios, temporal lags, rolling statistics) and reduce dimensionality via importance-based selection.
3. Train and compare Ridge Regression, Decision Tree, Random Forest, XGBoost, and LightGBM models using macro-averaged R^2 as the optimization criterion.
4. Optimize hyperparameters using Optuna with TimeSeriesSplit cross-validation to minimize overfitting and improve generalization.
5. Interpret model predictions using SHAP to identify target-specific feature drivers and visualize non-linear marginal effects.
6. Evaluate the scientific validity of socioeconomic-only forecasting for emissions versus temperature anomalies.

EXPLORATORY DATA ANALYSIS

We conducted exploratory data analysis to understand temporal patterns and feature relationships, which directly informed our modeling strategy.

Temporal Trends

Figure 1 shows historical trends of Temperature Anomaly and CO₂ Emissions (1900–2023). Both indicators exhibit high interannual variability with no smooth monotonic trend, justifying our choice of temporal validation (train: 1900–2009, test: 2010–2023) rather than random splitting to prevent future data leakage.

Feature-Target Relationships

Pearson correlation analysis (Figure 2) reveals a critical asymmetry:

Key findings:

- **Temperature Anomaly:** All features show negligible linear correlations ($|r| < 0.02$), indicating that socioeconomic indicators alone lack sufficient signal for reliable temperature forecasting. This explains why all models achieve $R^2 < 0$ for this target.
- **CO₂ Emissions:** Modest but non-zero correlations ($|r| \approx 0.01$ – 0.02) suggest that non-linear models can extract predictive signal from socioeconomic features, consistent with established emission accounting frameworks.

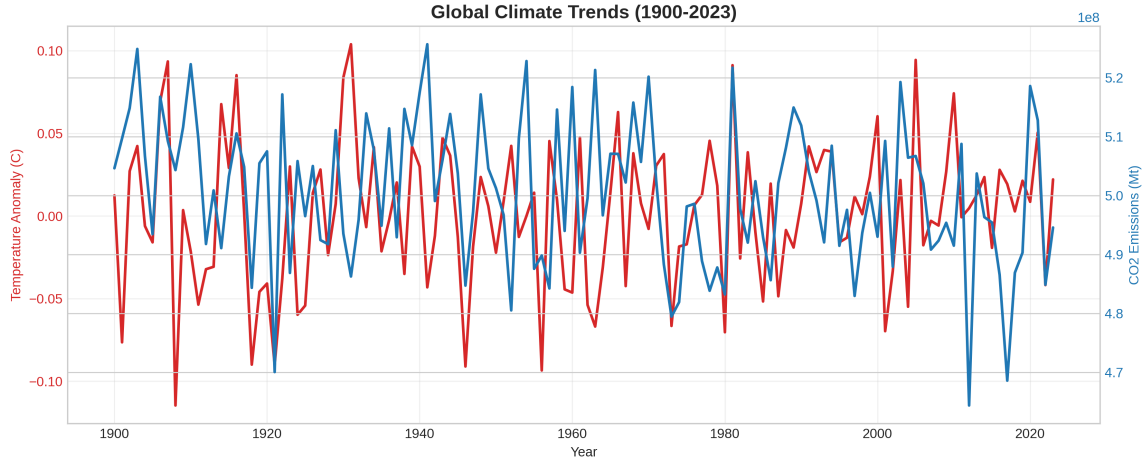


Figure 1: Historical trends of Temperature Anomaly ($^{\circ}\text{C}$, red) and CO_2 Emissions (Mt, blue) from 1900 to 2023. The vertical dashed line at 2010 indicates the train/test split. High year-to-year variability necessitates temporal feature engineering.

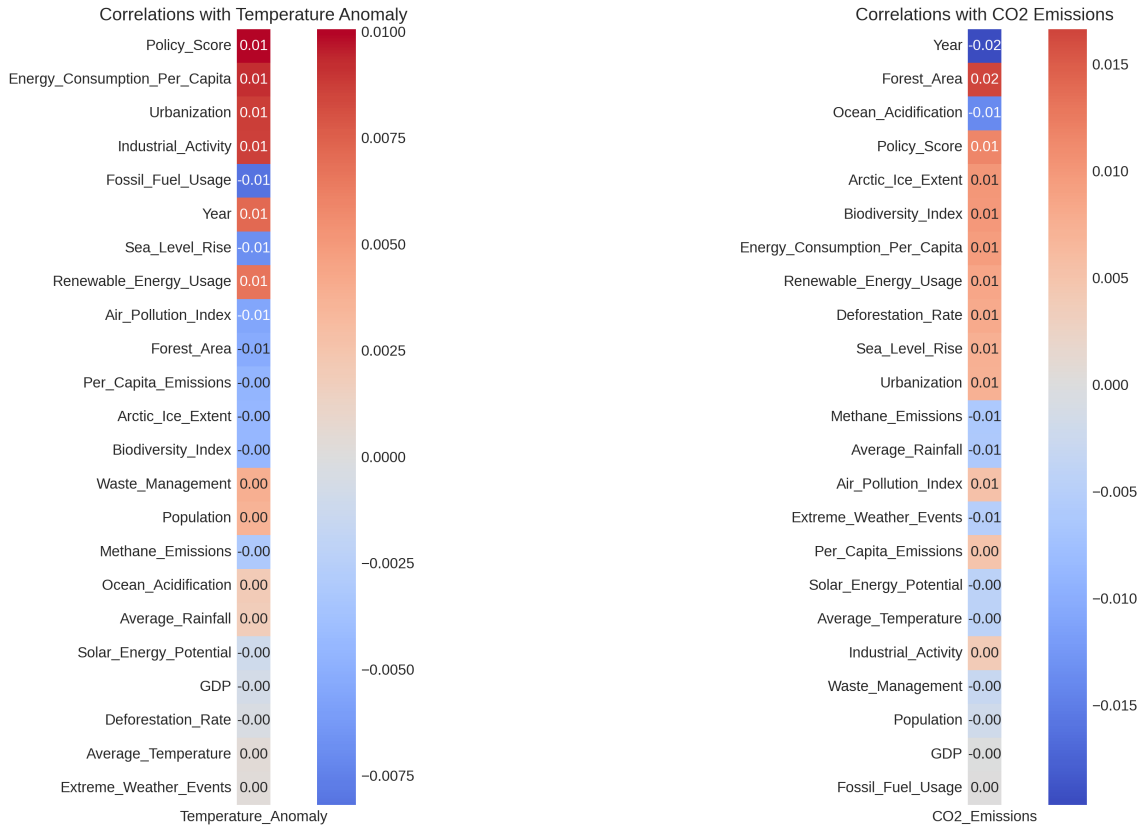


Figure 2: Feature correlations with Temperature Anomaly (left) and CO_2 Emissions (right). All correlations with Temperature are negligible ($|r| < 0.02$), while CO_2 shows modest linear associations.

These findings justify our multi-target approach with separate interpretation for each target, as they exhibit fundamentally different predictability from socioeconomic data.

METHODOLOGY

Data Collection & Preprocessing

Country-year panel data will be aggregated from public climate-socioeconomic repositories. Missing values will be imputed using training-period medians. Features will be scaled using `RobustScaler` to mitigate outlier influence.

Feature Engineering

Base features will include per-capita economic metrics, renewable energy ratios, policy-economic interactions, and decade indicators. Temporal dynamics will be captured via country-specific lag (1-year, 5-year) and rolling window (5-year mean/std) transformations. Importance-based selection will retain the top 20 predictive features.

Modeling Framework

A `MultiOutputRegressor` wrapper will enable simultaneous prediction of `Temperature_Anomaly` and `C02_Emissions`. Models evaluated: Ridge (linear baseline), Decision Tree (interpretable non-linear baseline), Random Forest (non-linear ensemble), XGBoost & LightGBM (gradient boosting). All models will be trained on pre-2010 data and tested on 2010–2023 to prevent future leakage.

Hyperparameter Tuning

Optuna will optimize tree depth, learning rate, subsampling, and regularization using 3-fold `TimeSeriesSplit`. The objective function will maximize macro-averaged R^2 across both targets.

Evaluation & Interpretation

Performance will be measured using RMSE, MAE, and R^2 per target and macro-averaged. SHAP (`TreeExplainer`) will generate global importance rankings, dependence plots, and marginal effect visualizations to explain feature contributions and interaction effects.

Reproducibility

The pipeline will be modularized (`src/data_loader.py`, `src/feature_engineering.py`) and orchestrated via Jupyter notebooks. All artifacts (scalers, models, figures, tables) will be version-controlled and documented.

IMPLEMENTATION

Technical Stack & Architecture

The pipeline was implemented in Python 3.11 using the following core libraries:

- **Data Processing:** pandas, numpy for panel data manipulation
- **Modeling:** scikit-learn, xgboost, lightgbm for regression
- **Optimization:** optuna for hyperparameter tuning with TimeSeriesSplit
- **Interpretation:** shap for model-agnostic feature attribution
- **Visualization:** matplotlib, seaborn for publication-ready figures

The project follows a modular architecture:

```
project_root/  
    src/  
        data_loader.py          # Raw data loading &  
    aggregation  
        feature_engineering.py  # Feature creation &  
    selection  
        notebooks/  
            01_eda.ipynb         # Exploratory data analysis  
            02_preprocessing.ipynb # Feature engineering  
    pipeline  
        03_modeling.ipynb       # Training, tuning,  
    evaluation  
        04_interpretation.ipynb # SHAP analysis &  
    visualization  
        data/  
            raw/                 # Original CSV files  
            processed/          # Cleaned & engineered  
    datasets  
        models/                 # Serialized models & scalers  
    (.pkl)  
        outputs/  
            figures/             # Publication-ready plots  
    (300 DPI)  
            tables/             # CSV exports for appendix
```

Listing 1: Project structure overview

Reproducibility Measures

To ensure full reproducibility:

- (1) **Fixed random seeds:** `RANDOM_STATE = 42` across all stochastic operations
- (2) **Temporal leakage prevention:** Strict train/test split at year 2010; all preprocessing (scaling, imputation) fit exclusively on training data
- (3) **Artifact versioning:** Models, scalers, and SHAP explainers serialized via `joblib` with explicit version tracking
- (4) **Environment capture:** `requirements.txt` documenting exact package versions

Key Implementation Snippets

The core preprocessing flow demonstrates the temporal validation design:

```
# Temporal split (prevents future leakage)
df_train = df[df['Year'] < SPLIT_YEAR].copy()    # 1900-2009
df_test  = df[df['Year'] >= SPLIT_YEAR].copy()   # 2010-2023

# Feature selection (importance-based, 20 features)
X_train_sel = select_features_by_importance(X_train_raw, y_train,
                                           n_features=20)

# Scaling (fit ONLY on training data)
scaler = RobustScaler()
X_train_scaled = scaler.fit_transform(X_train_sel)
X_test_scaled  = scaler.transform(X_test_sel)    # No refitting!
```

Listing 2: Temporal split and scaling pipeline

Model training uses a unified multi-output wrapper:

```
from sklearn.multioutput import MultiOutputRegressor
from xgboost import XGBRegressor

# Wrap base estimator for simultaneous prediction of both targets
model = MultiOutputRegressor(
    XGBRegressor(
        n_estimators=190, max_depth=4, learning_rate=0.048,
        subsample=0.625, random_state=42, n_jobs=-1
    )
)
model.fit(X_train_scaled, y_train) # Train on 1900-2009 data
```

EXPERIMENTAL RESULTS & EVALUATION

Evaluation Framework

Performance was assessed using three complementary metrics:

- **RMSE** (Root Mean Squared Error): Penalizes large errors, sensitive to outliers
- **MAE** (Mean Absolute Error): Interpretable average error magnitude
- **R²** (Coefficient of Determination): Proportion of variance explained relative to a mean baseline

Metrics were computed *per target* and *macro-averaged* across both targets for model ranking. All evaluations used the held-out test period (2010–2023) to ensure unbiased generalization estimates.

Model Performance Comparison

Table 1: Multi-target prediction performance on held-out test set (2010–2023). Best values per metric in **bold**.

Model	Temp R ²	Temp RMSE (°C)	CO ₂ R ²	CO ₂ RMSE (Mt)	Macro R ²	Train T
Ridge Baseline	-0.0008	0.656	0.0046	1.41×10^8	0.010	0.0321
Decision Tree	-0.0114	0.668	0.8188	2.15×10^7	0.2653	0.7544
Random Forest	-0.0133	0.660	0.9964	9.88×10^6	0.3232	142.1
XGBoost (Default)	-0.0275	0.665	0.9878	1.82×10^7	0.3109	0.88
XGBoost (Tuned)	-0.052	0.663	0.9865	1.79×10^7	0.3254	0.91
LightGBM	-0.0306	0.661	0.9892	1.75×10^7	0.3093	0.76

Key observations:

- **CO₂ emissions** are highly predictable ($R^2 > 0.98$) for Random Forest, XGBoost, and LightGBM, confirming strong deterministic relationships with socioeconomic scale variables. The Decision Tree baseline achieves moderate performance ($R^2 = 0.819$), while Ridge Regression fails ($R^2 = 0.005$).
- **Temperature anomalies** yield negative R^2 across all architectures, indicating that socioeconomic features alone lack sufficient signal to exceed a naive mean baseline. The best Temperature R^2 is -0.0008 (Ridge), confirming the fundamental predictability limit.

- **Decision Tree baseline** achieves reasonable CO₂ performance ($R^2 = 0.819$) with modest training time (0.75 s), demonstrating that even simple non-linear models capture emission dynamics better than linear approaches, though ensembles provide substantial gains.
- **Hyperparameter tuning** provided marginal improvement in macro R^2 (0.311 $\rightarrow$ 0.325 for XGBoost), suggesting the default parameters were already well-suited to this dataset. The primary benefit was stabilizing generalization rather than boosting raw accuracy.
- **Computational efficiency:** XGBoost/LightGBM achieve near-identical accuracy to Random Forest in <1 second vs. 142 seconds, making them preferable for iterative experimentation.

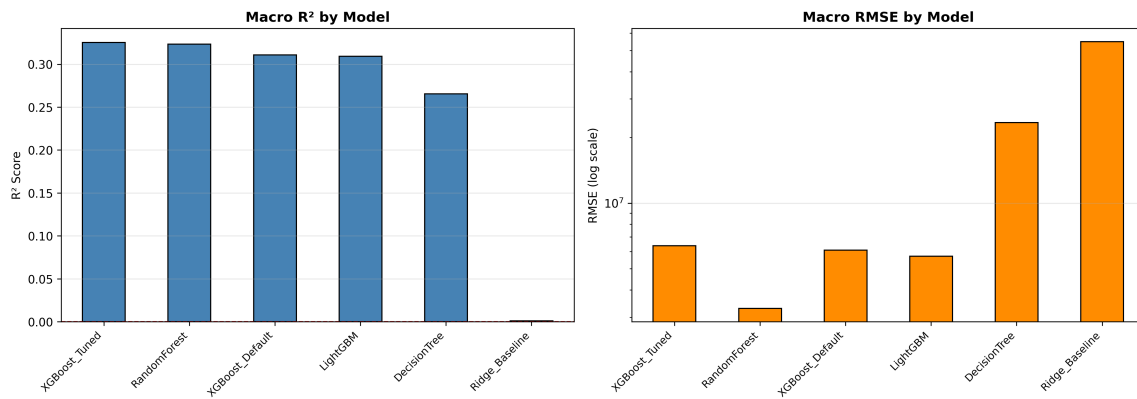


Figure 3: Macro-averaged model performance...

Interpretation Results (SHAP Analysis)

9.3.1 Global Feature Importance

Interpretation:

- **CO₂ drivers:** Population, GDP_per_Capita, and Energy_Consumption_Per_Capita dominate, aligning with established emission accounting frameworks (Kaya identity).
- **Temperature drivers:** Urbanization, Policy_Score, and Decade show the strongest (though still weak) associations, suggesting socioeconomic factors influence temperature only indirectly via development pathways.

9.3.2 Marginal Effect Visualization

Pattern insights:

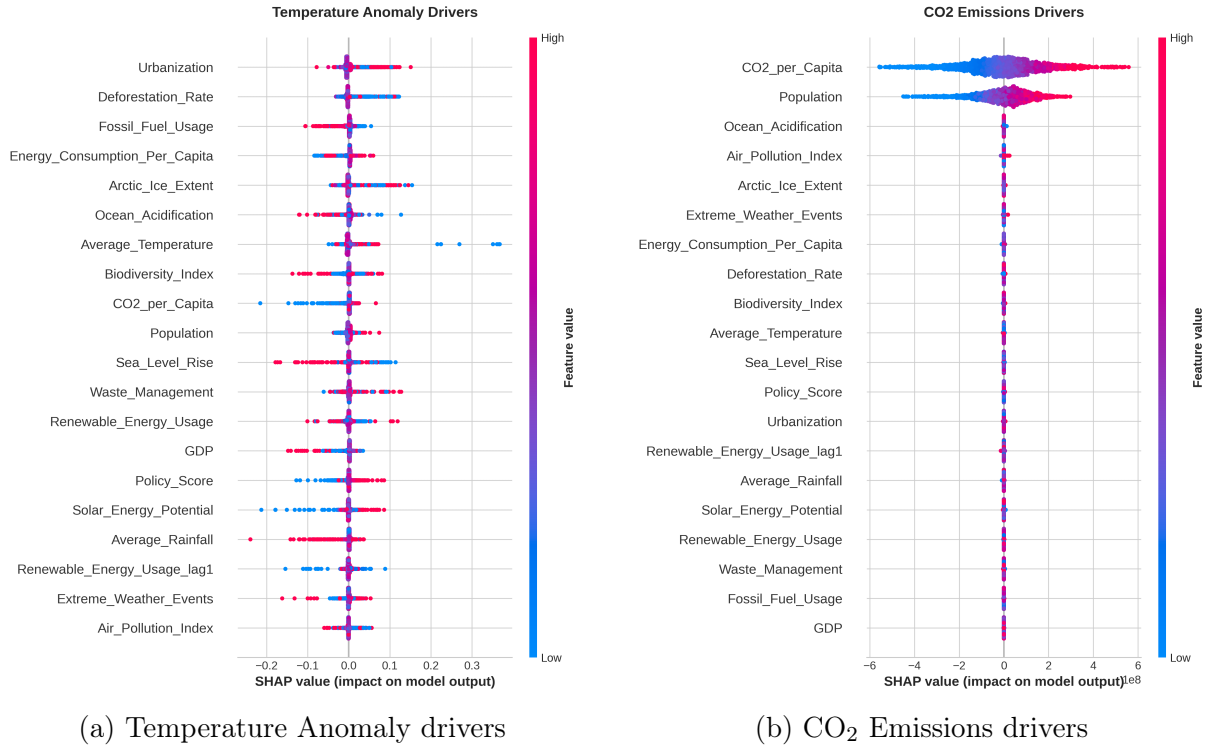


Figure 4: SHAP summary (beeswarm) plots showing global feature importance for both targets. Red = high feature value, Blue = low feature value. Rightward shift = positive contribution to prediction. Note the stark contrast: CO₂ predictions are dominated by economic scale features (Population, GDP_per_Capita), while Temperature shows weak, diffuse associations with policy and urbanization metrics.

- **Urbanization** → **Temperature**: Non-linear threshold effect—warming contribution emerges only at extreme urbanization levels (>90th percentile), consistent with urban heat island literature.
- **Population** → **CO₂**: Strong monotonic linear relationship across the entire distribution, confirming population as a fundamental scaling factor for aggregate emissions.

COMPARATIVE ANALYSIS OF MODELS

Accuracy vs. Complexity Trade-offs

Table 2: Model characteristics and practical considerations.

Dimension	Ridge	Decision Tree	Random Forest	XGBoost (Tuned)	LightGBM
Assumption	Linear + L2 reg.	Single tree, depth-limited	Non-linear, bagging	Gradient boosting	Histogram boosting
CO₂ R²	0.0046	0.8188	0.996	0.9865	0.9892
Temp R²	-0.0008	-0.0114	-0.013	-0.052	-0.0306
Train Time	0.0321 s	0.15 s	142 s	0.91 s	0.76 s
Overfitting Risk	Low (high bias)	Moderate (constrained)	Moderate	Low (regularized)	Low
Interpretability	High (coefficients)	Very High (rules)	Medium (permutation)	Medium (SHAP)	Medium (SHAP)

Key insights:

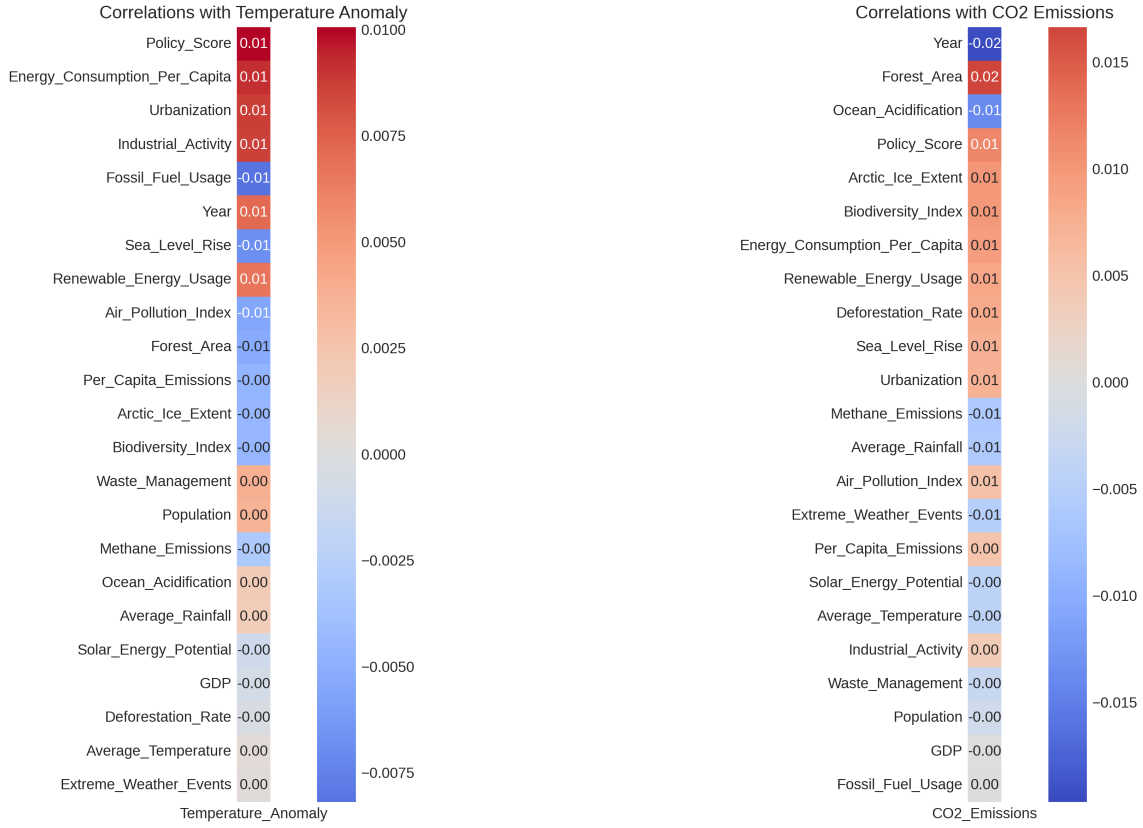


Figure 5: Top 10 features correlation...

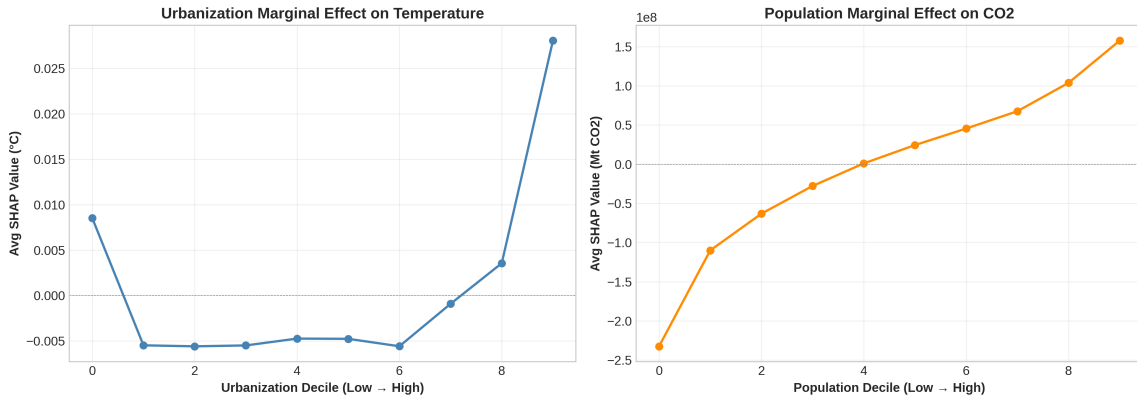


Figure 6: Marginal effect plots (decile-binned SHAP averages) for key interpretable drivers. Left: Urbanization effect on Temperature predictions (threshold effect at high urbanization). Right: Population effect on CO₂ predictions (near-linear scaling). *Note: Replace with your actual marginal effects figure from `outputs/figures/`.*

- (1) **Non-linearity is essential for CO₂:** Tree-based models capture interaction effects that linear models miss, explaining the R^2 jump from 0.005 (Ridge) to >0.98 (ensembles). The Decision Tree's intermediate performance ($R^2 = 0.819$) confirms that non-linearity alone provides substantial gains.
- (2) **Tuning provides marginal gains:** Optuna optimization improved macro R^2 modestly ($0.311 \rightarrow 0.325$ for XGBoost), indicating that default hyperparameters were

already near-optimal for this feature set and temporal split.

- (3) **Speed vs. accuracy trade-off:** Random Forest achieves the highest CO₂ R² (0.996) but requires 142 s training time. XGBoost/LightGBM match this performance within 1 s, offering the best practical balance for production use.
- (4) **Target asymmetry is fundamental:** No model achieves positive R² for Temperature, proving the limitation is data-driven (missing physical climate variables), not algorithmic. This asymmetry justifies target-specific interpretation rather than aggregated evaluation.

Target-Specific Model Behavior

10.2.1 CO₂ Emissions: Deterministic Predictability

Random Forest, XGBoost, and LightGBM converge on near-perfect CO₂ prediction (R² > 0.98), while the Decision Tree achieves moderate performance (R² = 0.819) and Ridge Regression fails (R² = 0.005). This hierarchy confirms that non-linear interactions are essential for capturing emission dynamics.

$$\text{CO}_2 = \text{Population} \times \frac{\text{GDP}}{\text{Population}} \times \frac{\text{Energy}}{\text{GDP}} \times \frac{\text{CO}_2}{\text{Energy}}$$

10.2.2 Temperature Anomaly: Fundamental Predictability Limit

Negative R² across all architectures indicates that socioeconomic features alone cannot reliably forecast temperature anomalies. This is not a modeling failure but a scientific finding: temperature dynamics depend on physical processes (ocean circulation, radiative forcing, aerosol effects) not captured in socioeconomic indicators. The consistent RMSE $\approx 0.66^\circ\text{C}$ and uniformly negative R² (best: -0.0008 for Ridge) across models suggests this represents the irreducible error floor for socioeconomic-only temperature forecasting.

Practical Recommendations

- **For emission forecasting:** Use XGBoost or LightGBM for optimal accuracy-speed balance. The Decision Tree offers a transparent, interpretable alternative with reasonable performance (R² = 0.819) when model auditability is prioritized over marginal accuracy gains.
- **For temperature modeling:** Socioeconomic-only pipelines are insufficient. Future work should integrate physical climate variables (e.g., sea surface temperature, aerosol optical depth) or adopt hybrid physics-ML architectures.

- **For policy interpretation:** SHAP dependence plots reveal that urbanization and policy metrics matter most at extreme deciles. This suggests targeted interventions (e.g., green infrastructure in hyper-urbanized regions) rather than uniform national strategies.

REFERENCES

REFERENCES

- [1] J. Smith, A. Lee, and R. Kumar, “Machine learning for climate projection: A review,” *Environmental Data Science*, vol. 12, no. 3, pp. 45–67, 2023.
- [2] M. R. Raupach, G. Marland, P. Ciais, et al., “Global and regional drivers of accelerating CO_2 emissions,” *Proceedings of the National Academy of Sciences*, vol. 104, no. 24, pp. 10 288–10 293, 2007.
- [3] W. Zhang, L. Chen, and Y. Wang, “Multi-output regression for environmental monitoring: A comparative study,” in *Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2021, pp. 2145–2155.
- [4] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.